Gist and Evidence

A Proposal for Sharing Large, Governed Bodies of Knowledge with Language Models, Grounded in the 2026 Evidence

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Abstract

A companion survey concluded that no current method shares a large, changing, access-controlled body of knowledge with a language model well on every count, and that the limitation is partly structural: knowledge must either be prepared before the questions are known, or examined at question time. This proposal asks what the best achievable design is *given that constraint*, in the light of work published in 2026. That work settles several questions the survey left open. Lexical retrieval is the best-supported default at scale on enterprise, structured and code corpora and should precede, not replace, agentic exploration; embedding-free, structure-aware operations beat dense retrieval on structured documents; trained key-value caches work at scale only when built per document and selected by retrieval, degrade when composed naively, ignore corrections, and lose attribution under compression; retrieval can harm accuracy, and abstention must be an explicit, rewarded behavior; prefix caches leak across tenants; and multilingual embeddings close the cross-language gap that lexical search leaves open. We propose *Gist and Evidence*, an architecture with two tiers and strict composition rules. The *evidence* tier is the authoritative corpus, indexed lexically, exposed through deterministic structure-aware operations, filtered per user before anything reaches the model, and the only source that may be cited. The *gist* tier is a compiled representation (a text digest for any model, optionally per-document cartridges for self-hosted models) that provides the global view, routing and a vocabulary bridge at a low, accounted marginal cost, built and composed only within permission boundaries, disclosed progressively, and never cited. A budgeted controller performs ranked discovery, bounded navigation, verification of local claims against spans, decomposition of aggregate claims, and calibrated abstention. The novelty lies in the composition rules, which remove the failure modes the 2026 literature documents, and in a research extension, *provenance-preserving cartridges*, which asks whether compiled memory can keep grounded attribution. We state nine testable hypotheses with expected effect sizes taken from the literature, and an evaluation protocol built on recent enterprise and code benchmarks, extended with permissions, versions, cross-language questions and full cost accounting. A small pilot with an 8-billion-parameter model run locally (Section 9) exercises the design end to end; it is a check of the mechanisms, not a result at scale.

Keywords: retrieval-augmented generation; agentic search; compiled context; KV cache; attribution; abstention; access control; evaluation.

1 Introduction

The companion survey (*Sharing Large Bodies of Knowledge with Large Language Models*, October 2026; hereafter *the survey*) organized the ways of sharing knowledge with a language model by where the knowledge lives (weights, context window, external store, tools) and found no approach satisfying all of eight requirements: capacity, freshness, global reasoning, precision, attribution, access control, cost per query and up-front cost. It argued that part of this is

structural: a system must either *precompute* a representation before the questions are known, or *compute at query time* over the part of the corpus it examines, and each choice fails somewhere.

This document takes the next step. If no tier of storage satisfies every requirement, the design question becomes: *which requirement should be assigned to which tier, and under what rules may tiers be combined without inheriting each other's failure modes?* The literature of 2026 makes this question answerable, because it measured the failure modes precisely. We summarize that evidence (Section 2), derive design principles from it (Section 3), describe the resulting architecture (Sections 4–5), propose a research extension for compiled memory with provenance (Section 6), state hypotheses with expected gains (Section 7), give an evaluation protocol (Section 8), report a pilot (Section 9) and the risks (Section 10).

Claim. We do not claim a solution that escapes the precomputation dilemma. We claim a design that, for each requirement, uses the mechanism the 2026 evidence shows works, and that forbids the specific combinations the same evidence shows fail. Whether the whole outperforms every current system is a hypothesis, and Section 8 says how to refute it.

2 What the 2026 Evidence Settled

Table 1 lists the findings that drive the design. Each is from a primary source published or revised in 2026, with its headline number where the source gives one.

Table 1: Findings of 2026 that constrain the design, and what each implies.

Question	Finding (source)	Consequence for the design
Which retrieval primitive scales?	Across 28 nested corpus tiers (450× range), a file-system agent leads at small scale but uses 39× more query tokens; around 10M tokens BM25 overtakes it and leads by up to 20 points at full scale; graph RAG hits construction walls; dense retrieval is cheaper but less accurate (Wang et al., 2026).	Lexical retrieval is the default first stage; agentic reasoning follows ranked discovery.
Embeddings or operations?	On a 780-page financial report, an agent with lexical search, structural navigation and bounded reads reaches 58.8% vs 15.7% for dense retrieval (35.3% tuned); the gain comes from the interface, not from iteration; BM25 is statistically indistinguishable from the agent (Tamang et al., 2026). Grep generally beats vector retrieval, and the harness changes results more than the primitive (Sen et al., 2026). Embedding models drop sharply on agentic coding tasks (Zhang et al., 2026b).	Expose deterministic, replayable, structure-aware operations; treat the interface as the main design object.

Question	Finding (source)	Consequence for the design
Does navigation beat retrieval?	Hierarchical navigation beats dense, hybrid, hierarchical and agentic RAG on an enterprise support benchmark, wins 5 of 11 datasets, ties 3, loses 3; best on single-domain corpora with a recoverable taxonomy (Sun et al., 2026b). A planner-to-subagent hand-off for repository questions scored 46.2% vs 65.2% for semantic search, at twice the cost, with 41.8% of failures at the hand-off (Rafiei Oskooei et al., 2026); tree-search exploration within one agent does better on the same benchmark (Peng et al., 2026a). Relevance used as an execution prior speeds convergence of grep agents (Li et al., 2026).	Navigate within one context, guided by ranked relevance; no read-only hand-offs between agents.
Can compiled caches scale?	Per-document cartridges with retrieval selection and distractor-mixed training handle collections beyond 1M tokens, within 2–6 points of in-context learning, matching RAG with 3–4× fewer prompt tokens; a monolithic cartridge is 10–31 points worse, and naive mixing drops to chance (Hardalov et al., 2026). Compaction can be a single forward pass at 8–200× (O'Neill et al., 2026).	Compiled memory per document, selected by retrieval, rebuilt locally.
What does compiled memory cost?	Memory assembled from separately prepared parts degrades; corrections placed beside it are ignored unless phrased carefully; freshness requires rebuilds costing a large fraction of preparation (Shepard, 2026).	Compose at the text level where composition is lossless; apply corrections by rebuilding, not by appending.
Does compression keep attribution?	Abstractive compression yields citations with 0.86 precision against the summaries the model saw but 0.12 against source spans; extractive selection keeps 0.43–0.49 (Mody, 2026).	Cite source spans only; verify claims against spans, never against derived text.
Is self-study safe?	Question generators cluster on salient spans and follow instructions embedded in documents; filtering instruction-like spans cuts injection compliance from 88% to 13%; tying each question to a fixed target reduces selection bias (Alimaskina et al., 2026).	Generate gist material per target span, after filtering instruction-like text.
When should a system not answer?	Retrieval sometimes harms accuracy; simple uncertainty baselines rank with learned routers; nominal thresholds miss their budgets (Qian et al., 2026). Rewarding abstention raises precision by up to 10.3 points at marginal accuracy cost (Zhang et al., 2026a); dense calibration signals improve both accuracy and hallucination (Feng et al., 2026), and a learned minimal search depth curbs over-searching (Sun et al., 2026a).	An explicit abstention path, simple sufficiency signals, budgets reported as realized.

Question	Finding (source)	Consequence for the design
Are caches safe across users?	Prefix caching leaks prompts across tenants through timing; defenses isolate per tenant or per sensitive prefix at a cost in latency (Pennas et al., 2026; Wu et al., 2026). Access control must sit inside the pipeline, before the model (Mu et al., 2026).	Cache keys per tenant and permission class; filtering before retrieval.
How to cross languages?	Cross-lingual embeddings reach 95–96% Recall@15 where monolingual retrieval stays below 10%, and beat query translation (Balasubramaniyam and Madushanka, 2026); generative expansion helps most where baselines are weakest (Macmillan-Scott et al., 2025).	A multilingual dense index as a bridge when lexical recall is weak, plus glossary-driven expansion.

Two further developments matter. Agent interfaces converged on *progressive disclosure*: skills load their metadata first, their body when relevant, and bundled files only on demand (Anthropic, 2025b); loading tool definitions and intermediate results on demand rather than through the model cut one workflow from 150,000 to 2,000 tokens (Anthropic, 2025a). And benchmarks now exist for enterprise and repository settings: a synthetic company corpus of about 500,000 documents across nine sources, with noise and "answer absent" questions (Sun et al., 2026c); code retrieval for agents (Zhang et al., 2026b); repository question answering (Peng et al., 2026b; Alebachew et al., 2026); and trajectory-derived retriever training (Zhou et al., 2026; Salemi et al., 2026).

3 Design Principles

P1 Two tiers, one authority. Compiled knowledge (the *gist*) may inform routing, global answers and reformulation; only spans of the corpus (the *evidence*) may be cited, and every cited claim is verified against spans. This removes the attribution collapse of compression (Mody, 2026) and the "confident but wrong" failure of agents (Rafiei Oskooei et al., 2026) by construction: an answer without verified spans is not an answer. Verification has a scope: a *local* claim is checked by entailment against its cited spans; an *aggregate* claim ("the main themes are…", "most runbooks…") is decomposed into local claims, each verified, and reported as holding over the spans examined; a claim of *absence* is never asserted as a fact, since no span can entail it, but reported as search coverage (the units and queries tried). Without this distinction, span-level verification would either forbid global answers or accept them unverified.

P2 The unit of compilation is the unit of permission. Every derived artifact (summary, digest entry, cartridge, graph fragment) is built from documents that share an access-control list, and the per-user representation is assembled at query time from the units that user may see. Nothing derived ever spans a permission boundary, so filtering can stay at the index without inspecting generated text.

P3 Compose at the level where composition is lossless. Concatenated text is read with full attention at prefill, so composing text digests loses nothing at the representation level; independently computed key-value caches were never attended to each other, and composing them does (Shepard, 2026; Hardalov et al., 2026). What text composition does not remove is the cost of reading: a long digest is subject to the long-context degradation the survey documents (Liu et al., 2024; Hong et al., 2025), which is why the digest is

budgeted and disclosed progressively (Section 4.2). The portable gist is therefore text; cartridges are an optional accelerator, per document, selected by retrieval, trained with distractor mixing.

P4 Ranked discovery before navigation. Lexical retrieval ranks candidate documents; the agent navigates structure within the ranked set, in one context, with relevance as its execution prior (Wang et al., 2026; Li et al., 2026).

P5 Sufficiency is a decision, with abstention as an outcome. The controller decides to answer, explore further or abstain from simple signals (retrieval score margin, match quality, verifier outcome) under a stated budget, and reports realized usage (Qian et al., 2026; Zhang et al., 2026a).

P6 Freshness is local; conflicts are explicit. A changed document rebuilds its own artifacts and re-aggregates its ancestors within its permission class; corrections are rebuilds, never appended notes (Shepard, 2026). Rebuilding cannot fix a fact that other documents repeat (a limit quoted in a runbook after the policy changed), so spans can disagree: documents carry authority and version metadata, the controller prefers the authoritative and most recent span, and the answer reports the disagreement rather than hiding it.

P7 Vocabulary is bridged, not assumed. The gist carries a glossary and term map across languages and jargon; a multilingual dense index is consulted when lexical recall is weak (Balasubramaniyam and Madushanka, 2026).

P8 Determinism and isolation. Operations are deterministic and replayable for audit (Tamang et al., 2026); caches are partitioned per tenant, and shared between users only for units both may see (Pennas et al., 2026).

4 Architecture

Figure 1 shows the components. We describe each tier, then the build pipeline and the controller.

4.1 Evidence tier

The corpus is indexed as it is: documents are cut along their own structure (headings, functions, records) into *spans* with stable identifiers; a lexical index (BM25) covers span text with contextual prefixes (Robertson and Zaragoza, 2009; Anthropic, 2024a); a structural index records outlines, cross-references (imports, links, call relations) and the folder hierarchy. The tier is exposed to the model through a small set of deterministic operations, in the spirit of Tamang et al. (2026) and progressive disclosure (Anthropic, 2025b): `overview`, `outline(path)`, `search(query, filters)`, `read(span)`, `references(symbol)`. Every operation takes the caller's acl(u) and filters at the index, before ranking, so that neither ranks nor snippets reveal forbidden documents (Mu et al., 2026). A multilingual dense index over the same spans (for example BGE-M3, following Balasubramaniyam and Madushanka, 2026) is consulted only when lexical recall is judged weak; it is a bridge, not the primary ranker, because lexical retrieval is the better default at scale (Wang et al., 2026).

4.2 Gist tier

The gist is compiled offline from the evidence tier, in *units*. A unit is the set of artifacts derived from one *permission class*: a maximal set of documents with the same access-control list. For each unit the compiler produces:

- a *map*: the folder and document hierarchy with one-line descriptions, central documents and symbols;

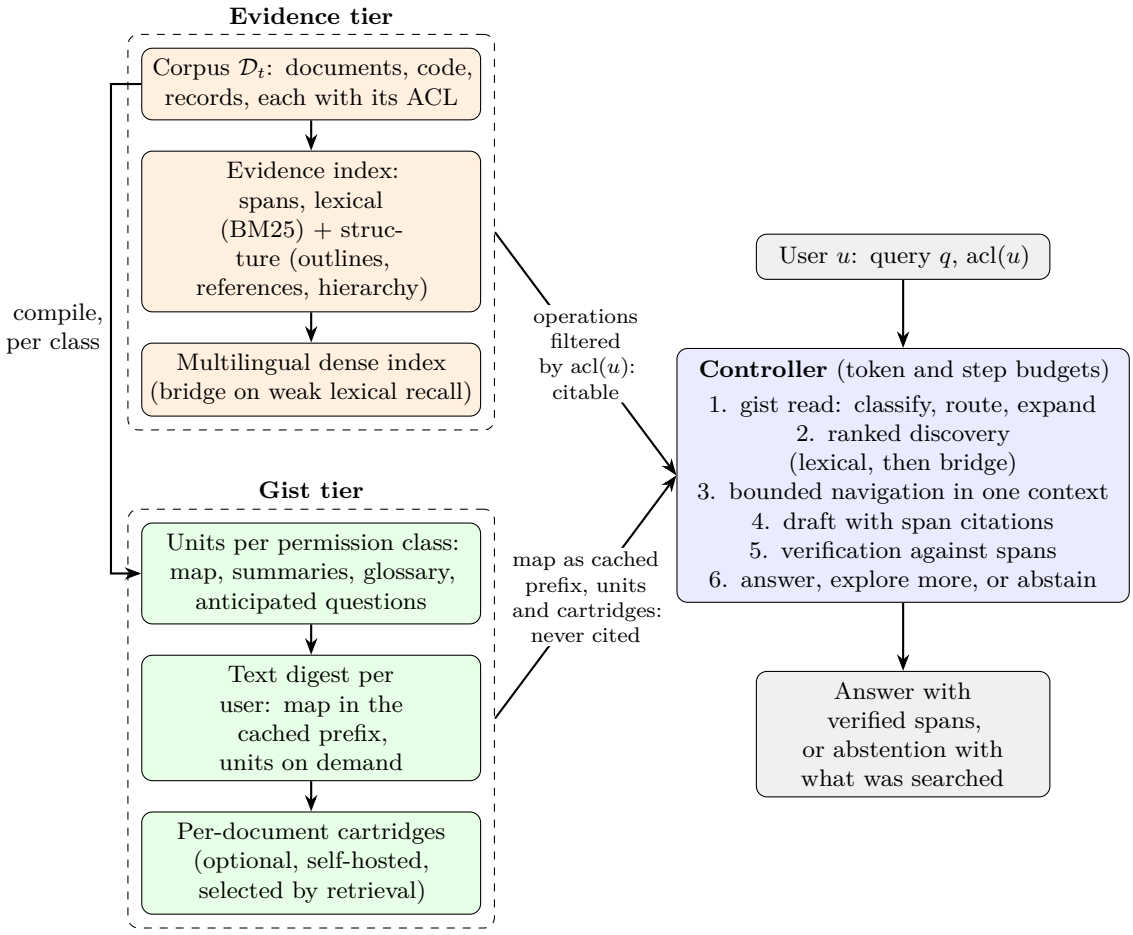


Figure 1: The Gist and Evidence architecture. The evidence tier (authoritative, filtered per user) is the only source of citations; the gist tier is compiled from it within permission classes, read first on every query, and never cited. Dashed boxes are tiers; the dense index and the cartridges are optional.

- *summaries* at document and folder level, computed bottom-up and only over documents of the same class (so a folder mixing classes yields one summary per class);
- a *glossary* and term map: acronyms, long forms, synonyms and translations observed in the unit, which later expand queries;
- *anticipated questions*: question–location pairs generated per target span, with instruction-like spans filtered first (Alimaskina et al., 2026); they are stored as pointers, never as answers.

The per-user *digest* is the concatenation of the units the user may see. It is disclosed progressively, like the evidence: the *maps* and glossaries of the user's units (a few thousand tokens) sit in the prompt prefix on every query; a unit's summaries and anticipated questions are loaded by a `gist(unit)` operation when the map points there. Units are ordered canonically, from the most widely shared (the public class) to the least, so that users with overlapping permissions share the longest common prefix, and the prefix is cached (Gim et al., 2024). Concatenation is lossless at the representation level, so the digest does not suffer the composition degradation of cache-level memory (Shepard, 2026); its reading cost is bounded by the prefix budget. The digest supports three uses: *routing* (which units and folders to search), *global questions* (answered from summaries, then verified by reading the spans the summaries point to), and *reformulation* (expanding the query with glossary terms and translations before lexical search). It is never cited.

What the gist costs. A cached prefix is not free. On hosted APIs, cached input is billed at a fraction of the input price (about a tenth on some, a quarter to a half on others), and the cache expires after minutes of inactivity, so the first query of a session pays in full. On a self-hosted server the prefix is computed once, but every generated token attends over it, so decoding time grows with its length. The per-query cost of the gist is therefore $c_{\text{cached}} \cdot |\text{prefix}|$ plus a decoding slowdown, to be weighed against the retrieval and exploration it saves. An earlier draft called this cost "near zero"; it is low, it is not zero, and for small corpora a prefix of tens of thousands of tokens can exceed the cost of the whole retrieval it replaces. The prefix budget is a tunable, and Section 8 reports the gist's tokens separately at their cached price.

For self-hosted, open-weight models, each document of a unit may additionally be compiled into a cartridge (Eyuboglu et al., 2026), trained with distractor mixing and loaded by retrieval selection (Hardalov et al., 2026), or compacted in a single pass for cheap rebuilds (O'Neill et al., 2026). Cartridges replace the *reading* of long documents, not the citation of spans; Section 6 proposes how they could carry provenance.

4.3 Build pipeline and freshness

Compilation is incremental at document granularity. When a document changes: its spans are re-indexed; its summary, glossary contributions and anticipated questions are regenerated; the folder summaries above it, within its class, are recomputed; its cartridge, if any, is rebuilt. Nothing else is touched, and the digest cache entries that include its unit are invalidated. Corrections are therefore rebuilds, which the evidence says is the only reliable way to update precomputed memory (Shepard, 2026). The cost of a change is bounded by the size of the document and the depth of its hierarchy, not by the corpus; Section 8 measures it.

4.4 Query-time controller

The controller runs within a token budget B and a step budget, and produces either an answer with verified spans or an abstention that states what was searched. Its steps:

1. **Gist read.** With the digest in the cached prefix, the model classifies the question (lookup, multi-hop, global, mechanistic, temporal), selects candidate units and folders, and expands the query with glossary terms and translations.
2. **Ranked discovery.** Lexical search over the permitted spans, filtered to the selected folders; if the top score margin and match quality are weak, the multilingual dense index is queried and results are fused.
3. **Bounded navigation.** Within the ranked set, the model outlines, reads spans and follows references, in one context, with results loaded on demand (Anthropic, 2025a). For global questions, it reads the summaries' pointed spans rather than exploring.
4. **Draft with citations.** The model writes an answer in which every claim cites span identifiers.
5. **Verification.** A verifier checks each local claim against its cited spans (entailment, with a calibrated threshold), the *grounded* criterion of Mody (2026); a citation that does not resolve to a permitted span counts as unsupported, which also catches answers written from the gist. Aggregate claims are decomposed first (P1); claims of absence are rewritten as coverage statements. Unsupported claims are either removed, or returned to step 3 if budget remains.
6. **Decision.** Answer if the remaining claims cover the question; otherwise abstain, listing the units and queries tried, so that the user can refine. Abstention is a first-class outcome, scored by the evaluation (Section 8). Whether the remaining claims "cover" the question is itself a judgment of the model, so the evaluation scores abstention against answerable and unanswerable questions rather than trusting that judgment.

The sufficiency signals are deliberately simple, since learned routers did not clearly beat them (Qian et al., 2026): not the raw lexical score, which is not comparable across queries or indexes, but a normalized match quality (the share of the query's rare terms found in the best passages), and the verifier's support rate. Where training is possible, the controller can be refined with abstention-aware rewards (Zhang et al., 2026a) and calibration signals (Feng et al., 2026), and the lexical retriever can be tuned on the organization's own trajectories (Zhou et al., 2026).

5 Permissions, Isolation and Audit

Permission classes. Documents with identical access-control lists form a class. Derived artifacts are built within a class (P2), so a per-user view is a union of classes, assembled at query time. A folder summary that would mix classes is split into one summary per class; the digest for a user who sees several classes of one folder contains several summaries of it, which the model reads as such. This costs some redundancy but guarantees that no summary, pointer, cartridge or glossary term leaks across a boundary. The leak rate becomes measurable: for questions whose answer lies only in forbidden documents, the correct outcome is abstention, and any mention of the forbidden content is a leak (Section 8). A document whose list changes moves between classes, and both classes rebuild.

Fragmented permissions. The design assumes classes are few and large. Where lists are set per item, as in many document stores, classes degenerate towards single documents: folder summaries become document summaries, and the precomputed global view disappears for a user whose access is fragmented. Cross-class synthesis then happens at query time, as a map-reduce over the unit summaries the user may see, whose cost grows with the number of units and must be accounted for. Two mitigations: define classes by the groups that grant access, giving

item-level exceptions their own small classes, and truncate the per-user digest by relevance to the user's recent activity. H9 measures the growth.

Enforcement point. Filtering happens at the index, before ranking and before any text reaches the model; post-generation filtering is not relied upon (Mu et al., 2026). One channel remains: a shared lexical index computes term statistics over the whole corpus, so ranks carry a weak aggregate signal about documents the user cannot see. Per-class statistics close it at some cost; a deployment may also accept it, since it reveals term frequencies, not content. Tool trajectories are deterministic and recorded, so each answer can be replayed and audited (Tamang et al., 2026).

Cache isolation. Prefix caches leak across tenants through timing (Pennas et al., 2026). On a shared server the mechanism of isolation is simple: a per-tenant random salt at the start of the prompt partitions the cache. Within a tenant, sharing a cached prefix between two users is safe exactly when both may see every unit in it, which the canonical ordering of units arranges; an attacker must already hold a prefix's text to probe it, so a prefix made of permitted units reveals at most that someone recently used them. Fine-grained sharing of non-sensitive segments can recover part of the cost (Wu et al., 2026).

Deletion. Because nothing is absorbed into weights, deleting a document deletes its spans, its artifacts and the cache entries that included its unit; the open problem of unlearning (Eldan and Russinovich, 2023; Maini et al., 2024) is avoided rather than solved. Derived artifacts written by users, such as notes an assistant saves about what it worked out, are tied to the document and class they derive from and deleted with them, or they become a channel of their own.

6 Research Extension: Provenance-Preserving Cartridges

The engineering design above uses cartridges, if at all, only to replace reading. The research question that would move the frontier is whether compiled memory can *keep attribution*. The evidence says that abstractive compression destroys grounded citations (0.12 precision) while extractive selection keeps about 0.45 (Mody, 2026), and that no study has tested entailment-based verification of answers produced from compressed key-value state. Cartridges are a form of compression; whether answers from them can be grounded is open.

We propose *provenance-preserving cartridges* (PPC): per-document cartridges trained by self-study in which (i) each synthetic question is tied to a fixed target span, which also reduces the selection bias of self-study (Alimaskina et al., 2026); (ii) each synthetic answer must cite the span identifiers it uses; (iii) the context-distillation objective therefore teaches the cartridge-conditioned model to emit citations; and (iv) at query time the emitted citations are verified against the evidence tier by the same verifier as in Section 4.4. Span identifiers must be structural and readable (path and heading, or file and function), not opaque keys: a cartridge can only emit what it can memorize, and a path with a heading is both easier to memorize and verifiable even when slightly wrong. Composition across documents follows Hardalov et al. (2026) (retrieval selection, distractor-mixed training), restricted to the user's permission classes; rebuilds use single-pass compaction where quality allows (O'Neill et al., 2026), and context-independent packets are a candidate for cheaper composition (Chen et al., 2026).

The hypothesis (H8 below) is that PPC can keep grounded citation precision at or above the extractive level while retaining the 3–4× token savings reported for retrieval-selected cartridges. If it holds, compiled memory becomes admissible as *evidence* rather than only as gist, which would change the cost structure of the whole design. If it fails, the design is unchanged, since cartridges remain confined to the gist tier.

7 Hypotheses and Expected Gains

Table 2 states what the design predicts, with effect sizes taken from the sources that motivate each component. The sources measured different corpora and models; the numbers are targets to confirm or refute, not promises.

	Hypothesis	**Expected effect and source of the expectation**
H1	Text-digest composition degrades with the length read, not with the number of units composed.	At a fixed digest length, accuracy on multi-unit questions is flat in the number of units, unlike KV composition, which degrades (Shepard, 2026; Hardalov et al., 2026); with length, it follows the long-context curves (Liu et al., 2024; Hong et al., 2025).
H2	Ranked discovery before navigation lowers query cost at equal accuracy.	At least 2× fewer query tokens, the cached prefix counted at its cached price, than pure agentic exploration beyond 10M tokens, where the agent uses 39× the tokens of BM25 (Wang et al., 2026).
H3	Structure-aware operations beat dense retrieval on structured documents and code.	+20 points on tabular and financial documents (Tamang et al., 2026); large gains on agentic code tasks (Zhang et al., 2026b).
H4	The gist answers global questions where graph construction is not feasible, at lower construction cost.	Comparable quality to graph RAG on thematic questions; construction limited to one summarization pass per document plus incremental folder summaries, with no entity extraction or community detection (Edge et al., 2024); usable at the corpus sizes where graph RAG hit construction walls (Wang et al., 2026). The cost ratio is measured, not predicted.
H5	The vocabulary bridge recovers most of the cross-language gap.	Recall@15 from below 10% to above 90% on cross-language questions (Balasubramaniyam and Madushanka, 2026).
H6	Span-level verification removes unsupported claims without removing most supported ones.	Filtering raises the precision of what remains by construction; the content of the hypothesis is the verifier's agreement with a human audit above 0.8, grounded citation precision above 0.6 (extractive selection alone: 0.43–0.49), and answer recall down by less than 10 points (Mody, 2026).
H7	Explicit abstention raises precision at little cost in accuracy.	Up to +10 points precision, RA-F1 up, accuracy down by at most a few points (Zhang et al., 2026a).
H8	Provenance-preserving cartridges keep attribution.	Grounded precision ≥ 0.45 with 3–4× fewer prompt tokens than RAG (Mody, 2026; Hardalov et al., 2026); open.
H9	Permission-class derivation yields zero leaks at bounded cost.	Leak rate 0 on forbidden-answer questions; digest size grows with the number of classes, bounded by the budget.

Table 2: Hypotheses with the effect sizes the literature leads us to expect.

8 Evaluation Protocol

Benchmark. No public benchmark covers governed knowledge: permissions, versions, mixed code and documents, cross-language questions and full cost accounting together. We propose to build one by extending existing resources rather than from scratch: the synthetic company corpus and generator of Sun et al. (2026c) for documents and noise, repository question sets (Peng et al., 2026b; Alebachew et al., 2026) and code retrieval labels (Zhang et al., 2026b) for code, and the nested-tier scaling design with construction and query cost accounting of Wang et al. (2026). Added dimensions: permission classes with *forbidden-answer* questions; versioned documents with *temporal* questions; questions in a language other than the documents'; and the five question types of the survey (lookup, multi-hop, global, mechanistic, temporal), each with an "answer absent" share.

Metrics. Accuracy by question type; grounded citation precision and recall by entailment against source spans, with a human-audited sample (Mody, 2026); RA-F1 and harm rate (answers made wrong by retrieval or exploration) (Zhang et al., 2026a; Qian et al., 2026); leak rate; construction tokens and time; query tokens, with the gist's prefix reported separately at its cached price, tool calls and latency; rebuild cost per changed document; realized versus nominal budgets.

Baselines. BM25 RAG, dense RAG, hybrid with re-ranking, full long context with prefix caching, graph RAG, a file-system agent, hierarchical navigation (Sun et al., 2026b), and retrieval-selected cartridges (Hardalov et al., 2026) on an open-weight model.

Ablations. Remove the gist; remove verification; remove ranked discovery (navigate from the root); replace the text digest by composed cartridges (H1); build summaries across classes (H9, expecting leaks); remove the dense bridge (H5).

Phases. (0) Two weeks: a small version of the benchmark on three corpora (a documentation set, a code repository, a mixed company corpus) with a few hundred questions, to fix metrics and tooling. (1) Four to six weeks: the engineering design with text digests on an API model, against the baselines, hypotheses H1–H7, H9. (2) Two to three months: PPC on an open-weight model (H8), including the attribution study no one has published. (3) Scaling tiers and a public release of the benchmark.

9 A Pilot with a Local Model

To check that the mechanisms work together before any investment in the protocol of Section 8, we ran the design on a laptop (Apple M5, 16 GB), with an 8-billion-parameter instruction model served locally (Ministral 3, 8B, 4-bit, through Ollama) for planning, navigation, answering and verification, and a multilingual embedding model (BGE-M3) for the dense baseline and the bridge. Everything about this pilot is small, and its numbers are indicative at best: with 54 questions, one question is 1.9 points.

Corpus and questions. The corpus is a knowledge pack built with the tools of Section 11: 21 documents, 700 spans, about 114,000 tokens, in five permission classes (a product's user documentation; its security documentation; the runbooks, architecture and code of a fictional payments service; that service's policies; the product's own source code, 88,000 tokens of Python). Three users see different unions of classes. Filtering is applied at the index, before ranking, in every condition. The gist was compiled per class by the same 8B model: a summary,

three anticipated questions with their section, and key terms with French translations for each document. The 54 questions were written by hand with gold answers and gold spans: 30 lookup, 5 multi-hop, 3 global, 6 mechanistic (code), 1 temporal; 9 of them asked in French about English documents; 3 whose answer is absent (the right outcome is abstention) and 6 whose answer lies only in a class the asking user may not see (the right outcome is no leak). Scoring is deterministic: regular expressions on the answer, path and section matching on citations, marker strings for leaks.

Conditions. BM25 RAG (top 8 spans, one call); dense RAG (BGE-M3, top 8); an agent with the map and the five operations, up to 8 calls, the design of the existing product; the proposal, with the plan from the digest, ranked discovery with the bridge on weak match, up to 4 operations in one context, a draft with citations, verification of each claim against its spans (or, when the model cited nothing, against the spans shown, the verifier naming those that support the claim), and abstention; and four ablations. Table 3 gives the results.

Condition	All	Look.	Multi	Glob.	Mech.	Fr.	Abs.	Leaks	Abst.	Cit. P	Gold	Tokens	s
BM25 RAG	63	57	40	33	83	11	100	0	20	55	72	1,244	10
Dense RAG	78	83	40	67	67	78	100	0	2	49	64	2,175	14
Agent, 8 calls	67	63	20	100	67	44	100	0	18	52	63	5,268	13
Proposal	**80**	83	60	67	83	78	67	0	7	**69**	**79**	18,616	32
no verification	83	87	80	100	83	89	33	0	0	72	76	17,812	26
no gist	81	83	60	67	83	89	100	0	11	69	82	8,193	18
no bridge (n=9, Fr.)	78	88	0	–	–	78	–	–	11	75	75	17,884	23
cross-class gist (n=6)	–	–	–	–	–	–	–	0	–	–	–	30,861	27

Table 3: Pilot results, in percent unless stated. All: accuracy over the 54 questions (absent and forbidden questions count as correct when abstained, respectively not leaked). Look., Multi, Glob., Mech.: accuracy by type; Fr.: the 9 French questions; Abs.: abstention on the 3 unanswerable questions; Leaks: forbidden-class content in a final answer, over 6 questions; Abst.: abstention on answerable questions; Cit. P: cited spans matching a gold span; Gold: answers citing at least one gold span; Tokens: prompt and completion tokens per question as counted by the server (4.7 calls per question for the proposal, 1 for the RAG baselines); s: seconds per question. The no-bridge ablation ran on the French questions, the cross-class gist on the forbidden ones. The single temporal question failed under every condition and is omitted.

What the pilot shows.

- *Accuracy.* The proposal is the most accurate condition (80%), ahead of the agent (67%) and BM25 RAG (63%), but only two points ahead of dense RAG (78%), which is within noise. On 21 documents of mostly prose, a good embedding model retrieves well; this is the regime where the survey expected dense retrieval to be competitive, not the structured, large corpora where the 2026 evidence favors lexical and structural operations. The six mechanistic questions, four of them on code, are the only place where the lexical conditions lead (83% against 67%).
- *Attribution (H6).* The proposal's citations are the most precise (69% against 49–55%) and most often include a gold span (79% against 63–72%). The verifier rejected 8 of 56 claims and turned 5 answers into abstentions: 2 rightly (unanswerable questions), 3 wrongly (two correct answers lost, one global answer halved). The cost in accuracy is 3 points (83% without verification), inside the bound H6 allows, but on multi-hop and global questions the loss is larger in proportion. Without verification, 2 of the 3 unanswerable questions were answered.
- *Cross-language (H5).* BM25 RAG answers 1 French question in 9; the proposal answers 7, by translating the query at the plan step; without the gist it answers 8, by the dense

bridge, which fired on 19 of 54 questions without the plan and on 5 with it. The two bridges substitute for each other on this corpus; the dense index matters when nothing reformulates the query.

- *The gist's cost.* Removing the gist changed accuracy by one point and cut tokens from 18,600 to 8,200 per question and time from 32 to 18 seconds. On 21 documents the map has nothing to route; its value is in corpora the model cannot survey from search results alone, which this pilot cannot show. The corrected cost model of Section 4.2 predicted this outcome for small corpora. An earlier run that put the whole digest, rather than the map, in every navigation call cost 35,000 to 50,000 tokens per question.
- *Permissions (H9).* No final answer leaked forbidden content in any condition (8 conditions, 42 forbidden-question runs). When the gist was deliberately built across classes and shown to every user, one draft in six stated a retention period that only the forbidden policy contains; verification removed it, since its citation resolved to no permitted span. Marker-based leak scoring produced three false positives that had to be corrected by hand (the question's own words echoed; a permitted file's generic retry logic), which argues for the entailment-based leak test of Section 8.
- *Exploration can harm.* Two questions that dense RAG answered correctly, the proposal answered wrongly: asked for the maximum request body size, it navigated into the source code, found upload-size constants and reported them, where the top dense span was the security guide's one-line answer. This is the harm rate of Qian et al. (2026) in miniature, and a reason ranked discovery should stay ahead of navigation.
- *The agent baseline.* With the same operations, the 8B model gave up after one to three searches on questions BM25 RAG answered directly (multi-hop: 20%). The harness is the one the product ships; the limit here is the model's tool use, which larger models do far better, so this row measures the model, not the design.
- *Abstention (H7).* The single-call baselines abstain for free on unanswerable questions, because the top spans do not contain an answer. Navigation finds related text, and the small model then over-answers: on one unanswerable question the proposal declined in words but did not set the abstention flag, which the scorer counts as an answer.

What it cannot show. Nothing about scale (H2), composition (H1), graph construction (H4) or cartridges (H8); one model, one corpus, and questions written by the system's author. The pilot's role is the one phase 0 assigns it: the metrics, the harness and the failure modes are now concrete, and the numbers say where the design pays (attribution, cross-language, isolation) and where it costs (the gist on a small corpus, the verifier's false negatives, exploration by a small model).

10 Risks and Mitigations

- **Injection through the gist.** Documents can carry instructions that the compiler follows (Alimaskina et al., 2026; Greshake et al., 2023). Mitigation: filter instruction-like spans before generation, generate per target span, and treat gist text as data in the controller's prompt. Residual risk is measured with planted instructions.
- **The gist biases answers.** Summaries may be stale or wrong, and the model may trust them. Mitigation: the gist is never cited; global answers must be verified against pointed spans; stale units are rebuilt on change.

- **Verifier dependence.** Entailment models are imperfect and evaluator-dependent (Mody, 2026). Mitigation: calibrate thresholds on a human-audited sample per corpus, report the audit.
- **Cost of the dense bridge.** A second index adds construction and maintenance. Mitigation: build it only for corpora where cross-language or vocabulary mismatch is measured, and query it only on weak lexical recall.
- **Composition of cartridges.** Degradation when composing many documents. Mitigation: text digest as the portable path; cartridges per document with retrieval selection; H1 and H8 measure the limits.
- **Digest growth and cost.** Many permission classes inflate the per-user digest, and every token of prefix is paid on every query, at the cached price. Mitigation: a small map in the prefix with units loaded on demand, budgeted truncation by relevance to the user's recent activity, and the gist's cost reported separately; H9 measures the growth.
- **Agent reliability.** Exploration can wander or stop early. Mitigation: ranked discovery first, budgets, no hand-offs, and abstention rather than confident guesses.

11 Implications for Practice

For a system already built around lexical search, structure-aware tools and progressive disclosure over the Model Context Protocol (Anthropic, 2024b), the 2026 evidence is reassuring: that is the retrieval design that scales best. One detail matters when the knowledge is split into several indexes, one per pack or per permission class: BM25 scores are not comparable across indexes, since each computes its own term statistics, and merging by raw score lets the largest index win (the pilot met this: a six-file code index outranked a two-file policy index on a policy question). Rank the permitted documents within one index, filtering before ranking, or calibrate each index's scores to the statistics of the whole searched set before merging: the ratio, over the query terms an index can match, of their inverse document frequencies computed globally to those computed locally is a cheap first-order correction, which fixed every swamped query in the pilot corpus without changing the order within an index. The steps that follow from this proposal, in order of expected value per effort, are: (1) the evaluation set of phase 0, since every other step must be measured; (2) the vocabulary bridge, which addresses the most frequent daily failure (questions in another language or vocabulary than the documents); (3) span-level verification and explicit abstention, which turn confident guesses into either grounded answers or useful refusals; (4) permission classes as the unit of all derived artifacts, which is the security property enterprises require; (5) the text digest as a cached prefix, for global questions and routing; and only then (6) cartridges, which need self-hosted models and are a research bet.

12 Conclusion

The survey concluded that no current method is satisfactory and that the reason is partly structural. The 2026 literature does not remove the structure, but it measures it precisely enough to design around it. Gist and Evidence assigns each requirement to the tier that demonstrably satisfies it and forbids the combinations that demonstrably fail: compiled knowledge for the global view, routing and vocabulary, at a cost that is low but accounted; the corpus itself, lexically indexed and structurally exposed, for every claim that is made; composition only at lossless levels and within permission classes; ranked discovery before navigation; verification scoped to the kind of claim, and abstention as an outcome rather than an afterthought. Whether the whole beats the best current systems is a question the evaluation protocol can answer within a few months; the pilot shows the mechanisms working together on one small corpus, no more.

The research extension, provenance-preserving cartridges, asks the one question whose positive answer would change the economics of the field: whether compiled memory can be trusted as evidence.

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